Results in Control and Optimization 14 (2024) 100365

Contents lists available at ScienceDirect

Results in Control and Optimization

journal homepage: www.elsevier.com/locate/rico

Enhancing daily stock trading with a novel fuzzy indicator: Performance analysis using Z-number based fuzzy TOPSIS method

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ARTICLE INFO

Keywords:
Fuzzy inference system
Fuzzy multi criteria decision making
Ranking performance
Technical indicator
Z-number

ABSTRACT

In the dynamic realm of financial markets, developing effective strategies for stock exchange transactions is paramount. This research addresses this critical need by introducing a pioneering indicator for daily stock trading, leveraging a robust fuzzy inference system (FIS). The indicator ingeniously integrates key technical indicators including Moving Average Convergence and Divergence (MACD), Relative Strength Index (RSI), Stochastic Oscillator (SO), and On-Balance-Volume (OBV). The FIS is meticulously constructed based on expert opinions and gleaned fuzzy rules. The fuzzified values then serve as inputs to the FIS, which in turn generates signals indicating optimal actions: buy, hold, or sell stocks. To validate the importance of the FIS, a carefully curated selection of stocks from the NASDAQ stock exchange is employed for experimentation. To prove the efficiency of the FIS, the technical indicators and the fuzzy risk-adjusted returns are considered as alternatives and criteria, respectively. A novel Z-number-based technique for order preference by similarity to the ideal solution (TOPSIS) method is used to rank the technical indicators and the FIS. The comparative results unequivocally demonstrate that the developed FIS surpasses existing indicators, yielding superior returns.

1. Introduction

Capital markets are complex systems influenced by various factors such as social, political, and psychological forces, leading to inherent non-linearities. Successful trading in these markets requires expertise, a deep understanding of market dynamics, and extensive practice. Nowadays, traders face the challenge of dealing with multiple variables, indicators, patterns, and other factors that influence financial decision-making within a limited time frame. Consequently, making investment decisions in the stock market can be a daunting task for many traders.

To address this challenge, investors rely on technical and quantitative analysis to determine the optimal timing for opening and closing orders. While conventional techniques have played a role in analyzing financial data, alternative approaches involving computational intelligence and soft computing methods have emerged. These strategies are better equipped to handle the dynamic, complex, non-linear, and volatile nature of stock markets [1–9].

This research is motivated by the need to develop a new soft computing-based technique for daily stock trading. By harnessing the power of soft computing, the proposed technique aims to address the intricacies of stock markets and improve decision-making processes. Through the application of computational intelligence and soft computing methods, this new approach seeks to enhance the analysis and interpretation of financial data, providing traders with more effective tools for navigating the complexities of the stock market.

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https://doi.org/10.1016/j.rico.2023.100365
Received 17 July 2023; Received in revised form 11 October 2023; Accepted 16 December 2023
Available online 19 December 2023
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Technical analysis is a methodology that aims to predict future stock prices based on past price and volume data of securities. Researchers such as Sahin and Ozbayoglu [10] have proposed a trend-normalized RSI indicator that utilizes a genetic algorithm to optimize its parameters. Da Costa et al. [11] have developed a trading system using technical indicators like Simple Moving Average (SMA), Exponential Moving Average (EMA), and MACD with the triple screen technique. Gunduz and Cataltepe [12] has created a model for daily stock forecasting in the Borsa Istanbul stock market. However, technical analysis heavily relies on historical data and the trader's ability to interpret trading signals for successful stock trading decisions.

To enhance the interpretability of trading signals, human knowledge, and experience play a crucial role. Incorporating human knowledge into the trading system is where fuzzy systems prove valuable. Fuzzy logic allows for the representation of variables as imprecise linguistic terms, enabling more precise identification of trading signals. Researchers such as Othman and Schneider [1] and Simutis [13] have highlighted the efficacy of fuzzy logic in predicting stock price movements, as it converts data into linguistic terms that are more meaningfully interpretable. Anbalagan and Uma Maheswari [14] have introduced the concept of a fuzzy meta graph for trading in the Indian stock market, employing a Fuzzy Inference System (FIS) and utilizing popular technical indicators such as RSI, MACD, SMA, and EMA to define rules within the FIS.

Researchers Nakano et al. [15] have introduced a fuzzy approach that combines multiple portfolio performance measurements to select the best-performing portfolio among several candidates. Lee and Jo [16] have proposed a fuzzy expert system based on a genetic algorithm (GA) approach for stock trading and market prediction. Chang et al. [17] have presented the support vector regression technique and a Takagi–Sugeno fuzzy rule-based model for forecasting stock price fluctuations, with testing conducted in the US stock market. Huang et al. [18] have utilized a fuzzy rule-based system to guide decision-making in stock trading, providing recommendations for buying and selling shares. Lam [19] has proposed a GA-based fuzzy expert system for stock trading. Dymova et al. [20] have integrated fuzzy logic and Dempster–Shafer theory methods for trading, demonstrating superior performance over existing techniques when tested with Warsaw Stock Index futures contracts. Tas and Gursoy [21] have developed a novel technical indicator using the Mamdani fuzzy inference system to identify buy, hold, and sell signals, showing that the fuzzy indicator outperforms other technical indicators. Lincy and John [22] have constructed an efficient framework based on multiple fuzzy inference systems for daily stock trading, with fundamental information serving as input. Their empirical results indicate that fuzzy inference systems outperform other technical indicators and existing models. Simutus [13] has proposed a stock trading system integrating fuzzy logic and evolutionary programming. Dourra and Siy [23] have developed a trading system that combines technical analysis and fuzzy logic, outperforming the buy-and-hold strategy. The research aims to incorporate popular technical indicators such as MACD, RSI, SO, and OBV into the indicator, categorize them as low, medium, or high, and fuzzify them using Gaussian membership functions. The FIS will be constructed based on expert opinions and fuzzy rules to generate signals indicating whether to buy, hold, or sell stocks. The performance of both the technical indicators and the FIS will be evaluated using the fuzzy analytic hierarchy process (FAHP) and the Z-number-based TOPSIS method to determine if the developed FIS outperforms existing indicators and provides better returns.

2. Novelty and contributions

Previous studies in stock trading analysis have predominantly relied on fundamental and technical indicators, often utilizing limited datasets. Notably, [21,22,24–27] demonstrated the superior performance of fuzzy inference systems (FIS) compared to traditional technical indicators. However, these studies primarily focused on assessing trading signal profitability, neglecting the critical aspect of risk associated with investment returns. In evaluating indicator performance, it is crucial to go beyond mere profit considerations. The risk-adjusted return, encompassing metrics such as the Sharpe ratio, Sortino ratio, Modigliani risk-adjusted performance (M^2), and average return, provides a comprehensive framework for decision-makers to compare high-risk high-return strategies with less risky options offering potentially lower returns.

In our present study, we address the limitations of prior research by incorporating risk-adjusted returns as a key criterion for indicator performance assessment. Unlike previous endeavors, which failed to consider this crucial aspect, we integrate a comprehensive evaluation framework that accounts for both profit and associated risk. Furthermore, while previous studies relied on fuzzy rule systems created by researchers based on their expertise, our approach introduces a new dimension. We sought guidance from two seasoned trading experts to develop a fuzzy inference system framework tailored to their collective experience and knowledge. This ensures a more realistic and contextually relevant model. In addition, we apply our proposed methodology to specific stocks within the NASDAQ share market. Recognizing the inherent uncertainty in financial markets, where returns are not deterministic and precise prediction is challenging, we leverage the effectiveness of fuzzy set theory to forecast stock returns. We represent these returns as triangular fuzzy numbers, allowing us to calculate fuzzy risk-adjusted returns for comparison against traditional indicators including RSI, MACD, SO, and OBV.

One of the distinctive features of our research is the introduction of Z-numbers, a novel mathematical framework that captures both fuzziness and information reliability. This approach surpasses the conventional type-1 and interval type-2 fuzzy sets in handling uncertainty. To the best of our knowledge, our study represents the first application of Z-number-based multi-criteria decision-making in stock trading performance analysis. This innovative method promises a more robust and nuanced assessment of indicator effectiveness. Our research presents a groundbreaking approach to stock trading analysis, combining risk-adjusted returns, fuzzy inference systems, and Z-number-based multi-criteria decision-making. By considering both profit and associated risk, and leveraging cutting-edge mathematical techniques, we aim to provide a more comprehensive and reliable framework for evaluating trading indicators.

The subsequent sections of this paper are structured as follows: Section 3 provides a detailed examination of the technical indicators used in stock trading. In Section 4, various types of fuzzy risk-adjusted returns are studied. The framework of the fuzzy indicator is presented and discussed in Section 5. Section 6 discusses the framework of the Z-FTOPSIS Method. Section 7 delves into the investigation of a case study problem. The comparative analysis of the proposed method is discussed in Section 8. Limitations of the proposed is given in Section 9. Finally, Section 10 concludes the paper and offers suggestions for future research.

3. Technical indicators

In the realm of security analysis, technical analysis plays a crucial role in identifying the strengths and weaknesses of a security by studying the patterns of price movements. This section aims to introduce the fundamental concept of technical indicators, such as RSI, MACD, SO, and OBV. These indicators serve as valuable tools for analyzing securities and provide insights into their price dynamics.

3.1. Moving Average Convergence and Divergence (MACD)

It is popularly known as MACD among the traders in the financial market, which is introduced by Murphy [28]. Subtract the longer EMA into shorter EMA, we get MACD. EMA of the price data is calculated as follows:

$$MACD(n) = \sum_{i=1}^{n} EMA_k(i) - \sum_{i=1}^{n} EMA_d(i) \tag{1}$$

$$EMA_n(i) = \gamma * \xi(i) + (1-\gamma) * EMA_n(i-1) \tag{2}$$

$$\gamma = \frac{2}{1+n} \tag{3}$$

where EMA_n is n-days exponential moving average, n is the number of days, and $\xi(i)$ is ith day closing price. In this study, k denotes a short period, and d represents a long period. To calculate EMA, we consider k=12 days and d=26 days. The buy and sell signals are generated as per signal line Eq. (4) [29].

$$Sign(n) = EMA_9(MACD(n)) \tag{4}$$

IF MACD(n) $\geq$ 0 and MACD(n-1) $\leq$ 0, THEN the Buy signal is generated.
IF MACD(n) $\leq$ 0 and MACD(n-1) $\geq$ 0, THEN the Sell signal is generated.

3.2. Relative Strength Index (RSI)

The RSI was developed by Welles [30] and defined as

$$RSI_t(n) = \frac{\sum_{i=0}^{n-1}(\xi_{t-i} - \xi_{t-i-1}) \mid (\xi_{t-i} > \xi_{t-i-1})}{\sum_{i=0}^{n-1} |\xi_{t-i} - \xi_{t-i-1}|} \times 100 \tag{5}$$

where RSI_t indicates the relative strength index at time t, ξ_t denotes the closing value of ith stock at time t, and n is the number of RSI periods. In this study, we consider n=14 days. RSI ranges from 0 to 100 after calculating RSI, and the buy and sells signals are generated as recommended in the original work by Wells [30] or as shown in Henderson [29] and the trading signal explains as follows:

IF RSI(n) $\geq$ 30 and RSI(n-1) $\leq$ 30, THEN Buy signal is generated.
IF RSI(n) $\geq$ 70 and RSI(n-1) $\leq$ 70, THEN Sell signal is generated.
IF the above conditions are not satisfied, THEN the Hold signal is generated.

3.3. Stochastic Oscillator (SO)

SO is developed by George Lane [31], which calculates the price changes over a period of time. Generally, it considers 14 days. The two signals of SO are %D and %K, which calculates as follows:

$$\%K = \left(\frac{\left(C_t - min\{\xi_t, \xi_{t-1}, \ldots, \xi_{t-n+1}\}\right)}{max\{Q_t, Q_{t-1}, \ldots, Q_{t-n+1}\} - min\{\xi_t, \xi_{t-1}, \ldots, \xi_{t-n+1}\}} \right) \times 100 \tag{6}$$

$\%D = 3-$ $days\ SMA\ of\ \%K$
$SO_t = 14-$ $days\ SMA\ of\ \%K$

where, C_t represents the closing price of today, ξ_t and Q_t represent the low and high price of the security during the given period, respectively. Buy or sell signal is generated when %D and %K crossovers. However, in this study, a buy signal is generated if the crossover takes place when SO is less than 20, and sell signals are generated if the crossover takes place when SO is more than 80, and the trading signal is explained as follows:

If$\%K < 20 \& \%D < 20 \& \%K > \%D$ THEN Buy signals generated
If$\%K > 80 \& \%D > 80 \& \%K < \%D$ THEN Sell signals generated
If the above conditions are not satisfied, THEN the Hold signal will be generated.

3.4. On Balance Volume (OBV)

OBV was introduced by Joe Granville [32], and measures buying and selling pressure as a cumulative value by considering positive and negative volume flow. OBV is calculated as follows:

$$OBV_t = \begin{cases} OBV_{t-1} + V_t, & C_t > C_{t-1} \\ OBV_{t-1} - V_t, & C_t < C_{t-1} \end{cases} \tag{7}$$

where C_t and V_t are the closing price and volume of the trading day t, respectively. The Buy signals are generated, when OBV is up trends, and the sell signal is generated when OBV is in a downtrend. The trading signal is explained as follows:

If OBV_{t+1} greater than $OBV_t \&\& OBV_{t+2}$ greater than $OBV_{t+1} \&\& OBV_{t+3}$ greater than $OBV_{t+2} \&\& OBV_{t+4}$ greater than OBV_{t+3} THEN Buy signals are generated.

If OBV_{t+1} less than $OBV_t \&\& OBV_{t+2}$ less than $OBV_{t+1} \&\& OBV_{t+3}$ less than $OBV_{t+2} \&\& OBV_{t+4}$less than OBV_{t+3} THEN Sell signals are generated.

4. Fuzzy risk adjusted return

Risk-adjusted return measures the level of risk associated with generating an investment return and is typically quantified as a numerical value or rating. In this research, the study incorporates fuzzy risk-adjusted returns as criteria, including the Fuzzy Sharpe ratio, Fuzzy Sortino ratio, Fuzzy Modigliani risk-adjusted performance (M2), and Fuzzy average returns. These fuzzy measures are utilized to calculate the respective risk-adjusted performance indicators in the following manner:

4.1. Fuzzy Sharpe ratio (FSR)

Sharpe ratio [33] has been ranked the portfolio based on the excess returns adjusted to the total risk. Here, the standard deviation of the return is considered as a risk.

$$\text{Sharpe ratio} = \frac{E(\tilde{R}_p) - R_f}{\tilde{\sigma}_r} \tag{8}$$

$E(\tilde{R}_p)$= Fuzzy average return of a technical indicator, R_f= Risk-free rate of return provided by bank interest rate, $\tilde{\sigma}_r$=Standard deviation of fuzzy return generated by the selected technical indicators. The higher the Sharpe ratio indicates the greater return of the risk taken.

4.2. Fuzzy Sortino ratio (FSoR)

Sortino and Price [34] proposed a new method to calculate the risk, which considers the deviation below the Minimal Acceptable Return (MAR). According to the Sharpe ratio, the risk is calculated as the deviation from the expected return. In contrast, the Sortino ratio viewed deviation above the expected return as a reward and deviation below the MAR as a risk. Hence, the Sortino ratio is calculated as follows:

$$\text{Sharpe ratio} = \frac{E(\tilde{R}_p) - R_f}{\tilde{\sigma}_d} \tag{9}$$

$E(\tilde{R}_p)$= Fuzzy average return of a technical indicator, R_f= Risk-free rate of return provided by bank interest rate, $\tilde{\sigma}_d$=Standard deviation of return generated by the selected technical indicators. The major defect of the Sharpe ratio is that it penalizes the extreme positive return and negative return in the same way. To avoid this issue, the Sortino ratio is considered the downside risk rather than the standard deviation. Here, also higher ratio indicates better performance and reveals a consistency in return.

4.3. Fuzzy Modigliani risk-adjusted performance (FMRAP)

Modigliani [35] proposed risk-adjusted performance for the comparison of performance only after properly adjusting the portfolio return for the risk-unmanaged market benchmark. The Fuzzy Modigliani risk-adjusted return is calculated as follows:

$$RAP = \frac{\tilde{\sigma_m}}{\tilde{\sigma_i}} E(\tilde{R_P}) - R_f + R_f \tag{10}$$

RAP= The risk-adjusted performance of the technical indicator. σ_m= The Standard deviation of unmanaged market index return, σ_i= The Standard deviation of return of the individual technical indicator. $E(R_p)$= Fuzzy average return of technical indicators, R_f= The risk-free asset return is the interest rate of the bank deposit. M^2 measures the risk premium for adjusted total risk. This ratio helps to compare the dissimilar assets. As M^2 uses capital market line it used to be considered as a superior measure. A higher ratio indicates better performance of the technical indicator.

4.4. Fuzzy average return (FAR)

The fuzzy average return generated by the technical indicator is also considered as a criterion for the ranking of technical indicator performance.

$$\text{Fuzzy average return} = \frac{\text{Fuzzy average return}}{\text{Total number of days}} \tag{11}$$

5. The proposed Fuzzy indicator framework

A Fuzzy Inference System (FIS) is a computational model that aims to replicate human intelligence by accommodating the inherent uncertainty and ambiguity present in real-world situations. It achieves this by employing a mathematical framework that incorporates "fuzziness", a concept that allows for the representation of imprecision and uncertainty. This approach is especially valuable in scenarios where traditional, deterministic logic may fall short. Central to the functionality of a FIS are the concepts of fuzzy membership functions and rules. These elements are instrumental in capturing and manipulating imprecise information. Fuzzy membership functions are mathematical constructs that assign degrees of membership to elements in a universe of discourse. They enable the modeling of linguistic variables, which are often subjective and context-dependent, in a quantitative manner.

The FIS encompasses two critical components: antecedents and consequents. Antecedents are the input variables, while consequents are the output variables. Each of these variables is associated with a set of fuzzy membership functions, which describe the degree of membership of a given input or output to various linguistic categories. These membership functions enable the FIS to represent imprecision in a way that aligns with human reasoning. The core of the FIS lies in the formulation of fuzzy IF-THEN rules. These rules express relationships between the antecedents and consequents in linguistic terms. By employing fuzzy logic, these rules accommodate the imprecision inherent in real-world situations. These rules are of paramount importance in decision-making processes, as they allow the system to navigate through complex, uncertain scenarios.

The FIS provides a robust foundation for creating systems capable of effective decision-making in situations marked by vagueness and inaccuracy. This is especially pertinent in domains where precise, deterministic models may not be suitable, such as in natural language processing, expert systems, and control systems for complex processes. Within the realm of fuzzy logic, two prominent types of FIS have emerged: the Mamdani system and the Sugeno system. The Mamdani system, also known as the IF-THEN rule-based system, employs linguistic variables to define the preconditions and post-conditions of rules. This system excels in handling complex, multi-variable scenarios. On the other hand, the Sugeno system adopts a more mathematical, function-based approach to define the relationship between inputs and outputs. It is particularly well-suited for applications requiring precise, mathematical modeling.

In the context of this study, the adoption of the rule-based multiple-inputs and single-output (MISO) framework indicates that the FIS considers multiple input variables to generate a single output. This configuration is well-suited for applications where various factors influence a singular outcome.A multiple-input and single-output fuzzy inference system characterized by the following definition:

$$\text{IF } X_1 \text{is } A_1 \text{ and } X_2 \text{ is } A_2 \ \cdots \ \text{AND / OR } X_k \text{ is } A_k \text{ THEN } z \text{ is } A_L$$

IF X_1 is A_1, ⋯, AND / OR X_k is A_k, where X_1 and X_k represent input variables, and A_1 and A_k represent classes defined on these variables. The postcondition, denoted as z, represents the output variable, and A_L represents the class defined on z. The antecedent of the rule indicates the degree to which the rule is applicable, while the conclusion assigns a membership function to the output variable. The output variable is assigned a numerical range from 0 to 100. A lower value implies a higher likelihood to sell the stock, whereas a higher value indicates a greater opportunity to acquire the stock.

5.1. Defuzzification

To obtain a crisp value from the fuzzy output, the process of defuzzification is employed. As the output variables in the fuzzy inference system (FIS) are inherently fuzzy, it is necessary to convert them into crisp values that can be understood by the system. Various defuzzification methods exist, and in this study, the center of the area method is utilized for defuzzifying the fuzzy output. The center of the area method is defined as follows:

$$Z_0 = \frac{\sum_{j=1}^{M} Z_j \mu_C(Z_j)}{\sum_{j=1}^{M} \mu_C(Z_j)} \tag{12}$$

where M denotes the number of quantization levels of control output C, Z_j represents the amount of control output at the quantization level j, $\mu_C(Z_j)$ indicates the membership value of the output variables. According to the calculation, a fuzzy value Z_0 is derived for a specific input value. The proximity of Z_0 to 100 (high end) indicates a strong buy signal for the stock, while its proximity to 0 (low end) suggests a strong sell signal. A trading signal is generated for each stock on a daily basis. Based on these day-by-day trading signals, the return of the stock is calculated.

Table 1
The fuzzy decision matrix.

	C_1	C_2	⋯	C_m
A_1	$(\chi_{11}^{(1)}, \chi_{11}^{(2)}, \chi_{11}^{(3)})$	$(\chi_{12}^{(1)}, \chi_{12}^{(2)}, \chi_{12}^{(3)})$	⋯	$(\chi_{1m}^{(1)}, \chi_{1m}^{(2)}, \chi_{1m}^{(3)})$
A_2	$(\chi_{21}^{(1)}, \chi_{21}^{(2)}, \chi_{21}^{(3)})$	$(\chi_{22}^{(1)}, \chi_{22}^{(2)}, \chi_{22}^{(3)})$	⋯	$(\chi_{2m}^{(1)}, \chi_{2m}^{(2)}, \chi_{2m}^{(3)})$
⋮	⋮	⋮	⋮	⋮
A_n	$(\chi_{n1}^{(1)}, \chi_{n1}^{(2)}, \chi_{n1}^{(3)})$	$(\chi_{n2}^{(1)}, \chi_{n2}^{(2)}, \chi_{n2}^{(3)})$	⋯	$(\chi_{nm}^{(1)}, \chi_{nm}^{(2)}, \chi_{nm}^{(3)})$

Table 2
The fuzzy numbers assigned to each linguistic term.

Linguistic terms	Fuzzy triangular number
Very Low (VL)	(0, 0, 0.25)
Low (L)	(0, 0.25, 0.5)
Average (A)	(0.25, 0.5, 0.75)
High (H)	(0.5, 0.75, 1)
Very High (VH)	(0.75, 1, 1)

6. Framework of Z-FTOPSIS method

In real-world situations, data often lack determinism due to incomplete or unavailable information, leading to vagueness or imprecision. To address this issue, an extension of the Z-fuzzy TOPSIS technique is employed to handle linguistic data. In multi-criteria decision-making (MCDM), the weights assigned to criteria are typically provided by experts using linguistic terms or determined through various methods. However, the weight of each criterion is uncertain and can vary depending on the method or expert. Therefore, ensuring the reliability of the criterion weight becomes crucial. Fuzzy numbers alone are insufficient to represent the varying weights of criteria. To address this, Z-numbers are utilized. The Z-fuzzy TOPSIS procedure is outlined below:

Step 1: Identify the alternatives and criteria.

Step 2: Construct the fuzzy decision matrix. In a fuzzy decision matrix, we assume that, each $\tilde{\chi_{ij}} = (\chi_{ij}^{(1)}, \chi_{ij}^{(2)}, \chi_{ij}^{(3)})$ is a triangular fuzzy number, which is given in Table 1.

Step 3: Compute the normalized fuzzy decision matrix as follows: For this, compute the $\alpha-$ cut for each fuzzy number which is defined as follows:

$$\tilde{\chi}_{ij} = \left[[\underline{\chi}_{ij}(\alpha)], [\overline{\chi}_{ij}(\alpha)]\right] = \left[\chi_{ij}^{(1)} + \alpha(\chi_{ij}^{(2)} - \chi_{ij}^{(1)}), \chi_{ij}^{(3)} - \alpha(\chi_{ij}^{(3)} - \chi_{ij}^{(2)})\right]$$

Therefore, each fuzzy number $\tilde{\chi_{ij}}$ has transformed to an interval, which is given by

$$[\underline{t}_{ij}(\alpha)] = \frac{[\underline{\chi}_{ij}(\alpha)]}{\sqrt{\sum_{i=1}^{n}\left([\underline{\chi}_{ij}(\alpha)]^2 + [\overline{\chi}_{ij}(\alpha)]^2\right)}}, \quad \text{and}$$

$$[\bar{t}_{ij}(\alpha)] = \frac{[\overline{\chi}_{ij}(\alpha)]}{\sqrt{\sum_{i=1}^{n}\left([\underline{\chi}_{ij}(\alpha)]^2 + [\overline{\chi}_{ij}(\alpha)]^2\right)}}, \quad i = 1, \ldots, m, \quad j = 1, \ldots, n$$

The normalized interval $\left[[\underline{t}_{ij}(\alpha)], [\bar{t}_{ij}(\alpha)]\right]$ transformed to a fuzzy number $\tilde{T}_{ij} = (t_{11}^{(1)}, t_{11}^{(2)}, t_{11}^{(3)})$ in the following manner. Substitute $\alpha = 1$, we get $t_{11}^{(2)}$. That is, $t_{11}^{(2)} = [\underline{t}_{ij}(\alpha = 1)] = [\bar{t}_{ij}(\alpha = 1)]$. Further, $t_{11}^{(1)}$ and $t_{11}^{(3)}$ are obtained when $\alpha = 0$. That is, $[\underline{t}_{ij}(\alpha = 0)] = t_{11}^{(1)}$ and $[\bar{t}_{ij}(\alpha = 0)] = t_{11}^{(3)}$. Therefor, $\tilde{T}_{ij}$ is the normalized triangular fuzzy number.

Step 4: Evaluate the criteria weight, which is discussed as follows: Z- numbers are very flexible to handle uncertain situations. It is represented by $Z = (\tilde{A}, \tilde{B})$, where $\tilde{A}(= (a_j^{(1)}, a_j^{(2)}, a_j^{(3)}), j = 1, 2, \ldots, n)$ is a triangular fuzzy number and $\tilde{B}$ is a reliability of $\tilde{A}$. In this article, the weights of the criteria are assumed to be a Z-number, which is collected from the experts. These linguistics terms of the reliability are transformed to fuzzy numbers which give in Table 2.

Step 5: Convert Z-number into fuzzy number by using Kang et al. [36] approach, which is denoted as $\tilde{z}$.

Step 6: Calculate the normalized weighted decision matrix. The matrix has been obtained by multiplying the normalized decision matrix by the weights associated with it as follows:

$$\tilde{p_{ij}} = \tilde{t_{ij}}.\tilde{z_j}$$

Step 7: Compute the positive ideal and negative ideal solution, which is obtained by

$$\tilde{Q}^+ = \left\{\tilde{p}_1^+, \tilde{p}_2^+, \ldots, \tilde{p}_n^+\right\}, \quad \tilde{Q}^- = \left\{\tilde{p}_1^-, \tilde{p}_2^-, \ldots, \tilde{p}_n^-\right\}$$

where $\tilde{p}_i^+ = (1, 0, 0)$ and $\tilde{p}_i^- = (0, 0, 0)$, $i = 1, 2, \ldots, n$ for each benefit or cost criteria.

Step 8: The distance between two fuzzy numbers can generally be used to calculate the separation of each alternative from the fuzzy positive ideal solution:

$$s_i^+ = \sum_{j=1}^{n} d(\tilde{p}_{ij}, \tilde{p}_{ij}^+), \quad i = 1, 2, \ldots, m$$

Similarly, the distance between the fuzzy negative ideal solution can be determined as follows:

$$s_i^- = \sum_{j=1}^{n} d(\tilde{p}_{ij}, \tilde{p}_{ij}^-), \quad i = 1, 2, \ldots, m$$

Step 9: Compute the relative closeness coefficient for each alternative, which is obtained by

$$R_i = \frac{s_i^-}{s_i^+ + s_i^-}, \quad i = 1, 2, \ldots, m$$

Step 10: Rank the alternative based on R_i value in descending order.

7. Case study

To assess the performance, historical stock price and volume data are employed to empirically test the selected technical indicators, namely RSI, MACD, SO, and OBV. The evaluation is conducted on a set of top 25 companies that are listed on the NASDAQ exchange.

The required historical data (from the calendar year 2015 to 2019) was collected from the website: http://www.nasdaq.com/Markets The company's name and its ticker are as follows: Apple Inc. (APPLE), Amazon.com.(AMZN), Alphabet Inc.(GOOGL), Adobe Inc.(ADBE), ASML Holding N.V. (ASML), AstraZeneca PLC. (AZN), Broadcom Inc. (AVGO) Berkshire Hathaway Inc.(BRKA), Berkshire Hathaway Inc. (BRK-B), Charter Communications, Inc. (CHTR), Comcast Corporation (CMCSA), Costco Wholesale Corporation (COST), Cisco Systems, Inc. (CSCO), Face book, Inc.(FB), Intel Corporation (INTC), Microsoft Corporation (MSFT), Netflix, Inc. (NFLX) NVIDIA Corporation (NVDA), PepsiCo, Inc. (PEP), QUALCOMM Incorporated (QCOM), Starbucks Corporation (SBUX), T-Mobile US, Inc.(TMUS), Tesla, Inc. (TSLA), Texas Instruments Incorporated (TXN), Walmart Inc. (WMT).

7.1. Input and output

In this analysis, we give careful consideration to four technical indicators: RSI, MACD, SO, and OBV. These indicators are treated as fuzzy variables of input, each contributing valuable insights to our assessment. To determine their values, we employ well-defined calculations based on established formulas (as indicated in Eqs. (1), (5–7) respectively) using historical stock price data. For each of these indicators, we define their membership functions using Gaussian distributions, as illustrated in Fig. 1. These membership functions serve to categorize the strength or significance of each indicator's value. Specifically:

7.1.1. MACD

- If the MACD value is above the signal line, as defined in Eq. (4), then the membership function is considered high.
- If the MACD value falls below the signal line, the membership function is considered low.

7.1.2. RSI

When it comes to the RSI, we employ three distinct membership functions:

- If the RSI value is below 30, the membership function is categorized as low.
- If the RSI value falls within the range of 30 to 70, the membership function is categorized as medium.
- If the RSI value exceeds 70, the membership function is categorized as high.

7.1.3. SO

The membership functions for the Stochastic Oscillator (SO) are as follows:

- If the SO value is below 20, the membership function is deemed low.
- If the SO value falls within the range of 20 to 80, the membership function is considered medium.
- If the SO value exceeds 80, the membership function is considered high.

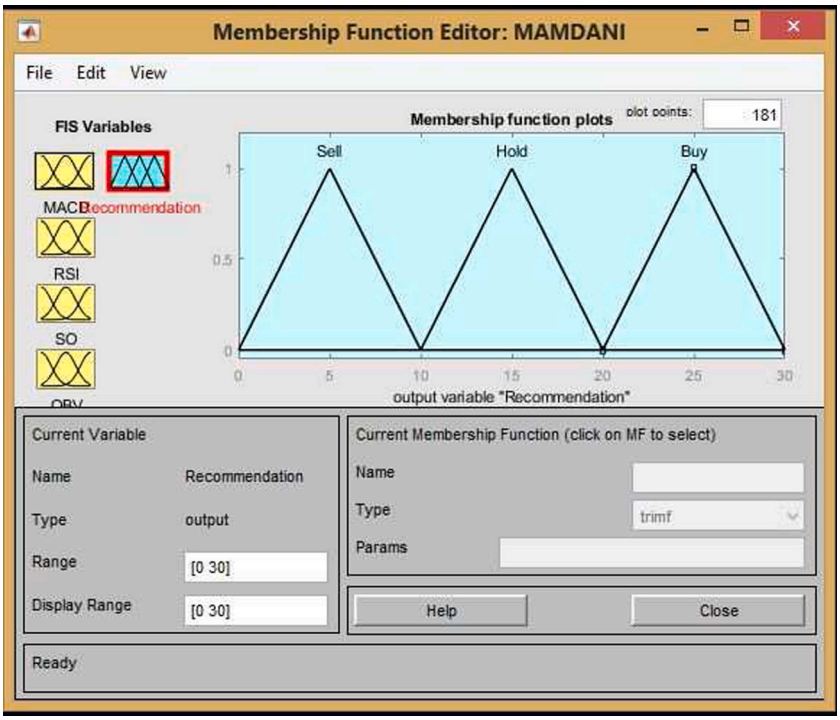


Fig. 1. Input and output membership function.

7.1.4. OBV

The membership function for the On-Balance Volume (OBV) is determined based on its trend:

- If the OBV exhibits an uptrend, the membership function is classified as high.
- If the OBV shows a downtrend, the membership function is categorized as low.

The technical indicators, namely RSI, MACD, SO, and OBV, are computed using historical data and then fuzzified using Gaussian membership functions. These indicators serve as input variables in the fuzzy inference system. The output variable is represented by a triangular membership function. Fig. 1 illustrates the membership functions for both the input and output variables. The fuzzy rules are formulated based on expert suggestions.

To develop a robust and effective fuzzy inference system tailored for stock market analysis, we leveraged the expertise of two seasoned trading professionals. Their extensive experience and hands-on knowledge in the trading field serve as a valuable foundation for crafting a system that not only demonstrates accuracy but also exhibits a high degree of contextual relevance. Within this framework, we identified four crucial input variables or parameters that play a pivotal role in evaluating a stock's performance.

The interaction of these four input variables forms the basis for establishing a set of fuzzy rules. The number of rules is contingent upon the linguistic labels assigned to each input variable. In this particular instance, if we assign three linguistic labels (e.g., "Low", "Medium", and "High") to each variable, and considering there are two variables with this labeling scheme, we arrive at a total of 3^2 (for the first two variables) multiplied by 2^2 (for the remaining two variables), resulting in a comprehensive set of 36 rules. These rules serve as the guiding principles for the fuzzy inference system, enabling it to navigate the complexity of stock market behavior with precision and clarity. These formulated rules are compiled and presented in Table 3. This table serves as a tangible representation of the expert-derived rules, offering a clear and organized view of the logic that underpins the fuzzy inference system. Each entry in the table corresponds to a specific rule, providing a transparent and interpretable framework for understanding how the input variables contribute to the final assessment of stock performance.

7.2. FIS result

The proposed framework has been implemented in the Matlab fuzzy logic toolbox. Figs. 1 and 2 display the interfaces of the input and output variables' membership functions, while Fig. 3 showcases the rule viewer. The input variables utilized in the framework are RSI, MACD, SO, and OBV, which serve as technical indicators. The rules used in the system are derived from technical trading experts. For instance, when the input values for MACD, RSI, SO, and OBV are 10, 100, 100, and 10, respectively, the fuzzy system generates an output value of 5.35, indicating a sell signal, as depicted in Fig. 3. Similarly, buy and hold signals are generated

Table 3
The rules of classification corresponding to Buy FIS are.

		IF		THEN
MACD	RSI	SO	OBV	Strength
Low	Low	Low	High	Buy
High	Low	Low	Low	Sell
High	Medium	Low	High	Buy
Low	Low	Low	High	Hold
Low	Low	High	High	Sell
high	Medium	High	Low	Buy

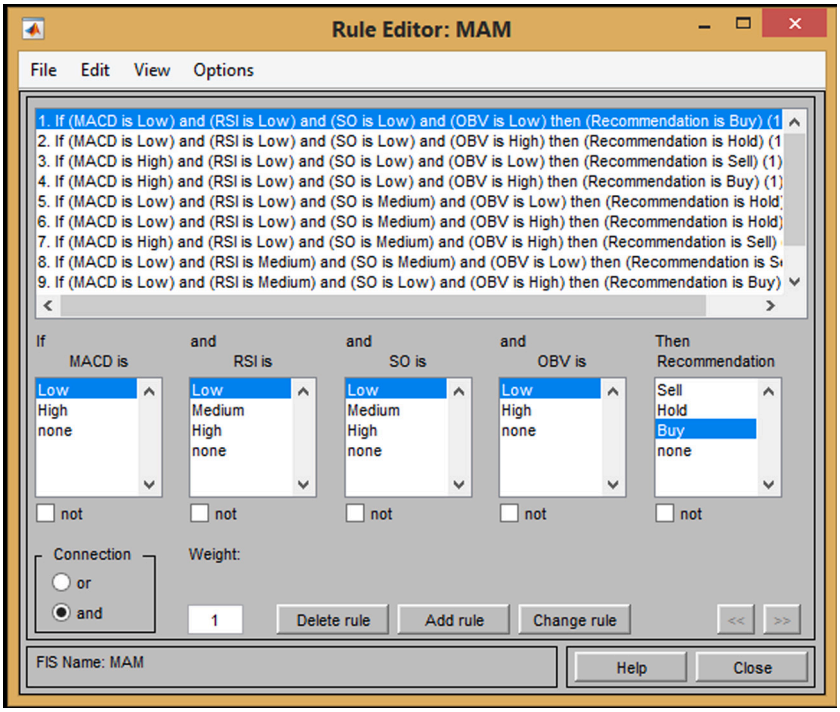


Fig. 2. The rule editor.

based on the input data. After obtaining the trading signals for each day using the fuzzy inference, the returns are computed. The effectiveness of the developed framework is evaluated through statistical methods and the Z-number-based fuzzy TOPSIS method.

8. Comparative results

In this section, we compare the proposed FIS framework over the technical indicators such as MACD, RSI, SO, and OBV, through statistical analysis, and the Z-number-based fuzzy TOPSIS method.

8.1. Statistical analysis

In this section, a comparative analysis between the FIS framework and the technical indicators (MACD, RSI, SO, OBV, and Buy-Hold Strategy) is conducted using statistical methods. Trading signals (buy, sell, hold) are generated using MACD, RSI, SO, OBV, and BH. Based on these signals, the day-by-day returns of each stock are calculated. Table 4 presents the average return and standard deviation for each stock. Additionally, Table 5 displays the average return, standard deviation, minimum value, and maximum value of returns for each indicator.

From Table 5, the average returns for RSI, MACD, SO, OBV, FIS, and BH are found to be 0.1243, −0.0082, 0.0004, 0.2632, 0.2653, and 0.0656, respectively, indicating that FIS provides the highest average return. The minimum return values for RSI, MACD, SO, OBV, FIS, and BH are 0.0242, −0.1179, −0.1134, 0.1427, 0.1599, and −0.0336, respectively. This suggests that FIS yields the highest minimum return compared to other indicators. Similarly, FIS demonstrates the highest maximum return. Furthermore, FIS exhibits the lowest standard deviation, indicating that it offers the least amount of risk.

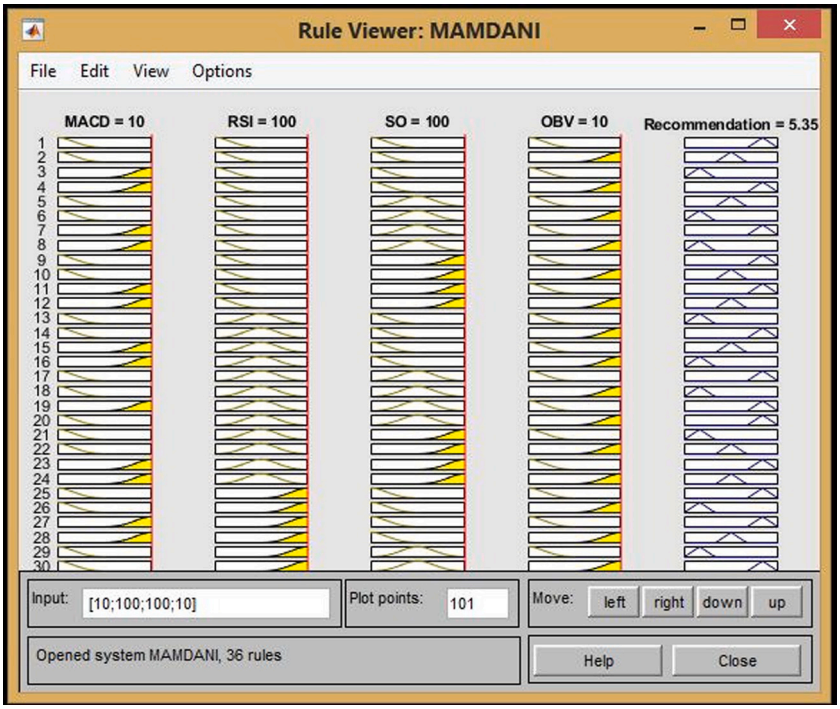


Fig. 3. The rule viewer.

Table 4
Risk and return of sample firm for selected technical trading indicators.

Stock	RSI		MACD		SO		OBV		FI	
no.	Return	Risk	Return	Risk	Return	Risk	Return	Risk	Return	Risk
1	0.1086	0.4487	0.1147	0.5159	−0.0275	0.5025	0.1716	0.4619	0.2186	0.4934
2	0.3438	0.3543	−0.1098	0.5213	0.0004	0.4778	0.2157	0.3865	0.2179	0.3848
3	0.1705	0.4872	−0.0356	0.5360	0.0744	0.4201	0.2609	0.3876	0.2631	0.3904
4	0.1560	0.4341	−0.0407	0.4603	−0.1097	0.5366	0.2506	0.4496	0.2492	0.4482
5	0.0876	0.4707	−0.0378	0.4925	0.0292	0.5313	0.2723	0.3732	0.2765	0.3746
6	0.0547	0.4810	0.1658	0.4736	−0.0681	0.5314	0.1706	0.4558	0.1693	0.4546
7	0.0935	0.4606	−0.0094	0.5408	0.0492	0.4743	0.3362	0.3584	0.3376	0.3583
8	0.1417	0.4923	−0.0033	0.5120	0.0080	0.5225	0.3015	0.3633	0.3002	0.3631
9	0.1309	0.5205	−0.0323	0.3473	0.1187	0.4736	0.2846	0.3989	0.2825	0.3953
10	0.0407	0.4960	−0.0483	0.4216	0.0415	0.4707	0.2978	0.4180	0.2822	0.4131
11	0.1218	0.4895	0.0995	0.4571	0.0920	0.4892	0.2975	0.3804	0.2972	0.3844
12	0.0242	0.4645	0.1231	0.5414	−0.0221	0.5117	0.3325	0.4048	0.3177	0.3905
13	0.1893	0.4198	−0.1005	0.4876	0.0737	0.4659	0.2274	0.4101	0.2111	0.3829
14	0.1246	0.4785	0.1050	0.5021	0.0411	0.4688	0.2824	0.4148	0.2824	0.4148
15	0.0451	0.4750	−0.0254	0.4255	−0.1134	0.4398	0.2799	0.3683	0.3099	0.3915
16	0.1782	0.4529	−0.0549	0.4777	0.0037	0.5272	0.2611	0.3965	0.2588	0.3976
17	0.1417	0.4001	−0.0103	0.5704	0.0379	0.5104	0.2315	0.3717	0.2211	0.3622
18	0.2126	0.4588	−0.1090	0.4225	−0.0708	0.4736	0.2831	0.3460	0.3199	0.3738
19	0.1257	0.5006	−0.0339	0.5412	−0.0098	0.4969	0.3148	0.3875	0.2945	0.3842
20	0.0489	0.4852	0.0843	0.4594	−0.0111	0.5175	0.2084	0.3948	0.2017	0.3999
21	0.1844	0.3874	−0.0108	0.4905	0.0524	0.4436	0.2800	0.3826	0.2676	0.3667
22	0.0399	0.4400	−0.1082	0.4852	−0.0361	0.4308	0.1427	0.3695	0.1599	0.3837
23	0.1420	0.4788	−0.1179	0.4533	−0.0474	0.4631	0.2402	0.4110	0.2608	0.4251
24	0.0770	0.5185	−0.0319	0.5303	−0.0388	0.5074	0.3082	0.3815	0.3137	0.3862
25	0.1242	0.4783	0.0228	0.5764	−0.0568	0.5412	0.3275	0.4246	0.3195	0.4231

Table 5
Descriptive statistical analysis: FI vs RSI, MACD, SO, OBV.

	RSI	MACD	SO	OBV	FI
Mean	0.1243	−0.0082	0.0004	0.2632	0.2653
Standard Deviation	0.0697	0.0807	0.0608	0.0518	0.0484
Median	0.1246	−0.0319	0.0004	0.2799	0.2765
Kurtosis	2.7848	−0.4420	−0.5580	−0.0179	−0.3800
Skewness	1.1271	0.6410	−0.0494	−0.7436	−0.6421
Minimum	0.0242	−0.1179	−0.1134	0.1427	0.1599
Maximum	0.3438	0.1658	0.1187	0.3362	0.3376

Table 6
Comparison of FIS with other selected TI.

	RSI Vs FI	MACD Vs FI	SO Vs FI	OBV Vs FI	FI Vs BH
t-Stat	−8.11606	−14.3936	−17.4942	−0.6465	−17.9400
P(T<=t) one-tail	1.22×10^{-8}	1.33×10^{-13}	1.82×10^{-15}	0.2620	1.04×10^{-15}
t Critical one-tile	1.7109	1.7109	1.7109	1.7109	1.7109
P(T<=t) two-tail	0.00	0.00	0.00	0.52	0.00
t Critical two-tail	2.06	2.06	2.06	2.06	2.06

Table 7
Fuzzy decision matrix.

	FSR	FSoR	FMR	FAR
RSI	(5.3183, 6.7547, 9.9043)	(5.7471, 9.1186, 16.5117)	(5.1969, 6.0115, 7.7978)	(5.0242, 5.1243, 5.3438)
MACD	(3.5131, 4.8734, 7.0292)	(0.4328, 4.6113, 11.2331)	(3.8893, 4.9229, 6.5607)	(4.8821, 4.9918, 5.1657)
SO	(3.1019, 4.9734, 6.9185)	(1.1837, 4.9465, 8.8573)	(4.1945, 5.0028, 5.8430)	(4.8866, 5.0004, 5.1187)
OBV	(7.7152, 10.0399, 11.4491)	(12.7114, 19.3137, 23.3159)	(5.8631, 6.5915, 7.0331)	(5.1427, 5.2631, 5.3361)
FI	(8.2598, 10.4344, 11.9257)	(8.5253, 10.8768, 12.4896)	(5.9049, 6.5009, 6.9096)	(5.1599, 5.2653, 5.3376)

Table 8
The fuzzy decision matrix and weights of criteria by linguistic terms.

	FSR	FSoR	FMR	FAR
RSI	(5.3183, 6.7547, 9.9043)	(5.7471, 9.1186, 16.5117)	(5.1969, 6.0115, 7.7978)	(5.0242, 5.1243, 5.3438)
MACD	(3.5131, 4.8734, 7.0292)	(0.4328, 4.6113, 11.2331)	(3.8893, 4.9229, 6.5607)	(4.8821, 4.9918, 5.1657)
SO	(3.1019, 4.9734, 6.9185)	(1.1837, 4.9465, 8.8573)	(4.1945, 5.0028, 5.8430)	(4.8866, 5.0004, 5.1187)
OBV	(7.7152, 10.0399, 11.4491)	(12.7114, 19.3137, 23.3159)	(5.8631, 6.5915, 7.0331)	(5.1427, 5.2631, 5.3361)
FI	(8.2598, 10.4344, 11.9257)	(8.5253, 10.8768, 12.4896)	(5.9049, 6.5009, 6.9096)	(5.1599, 5.2653, 5.3376)
weight	((0.3081,0.4216, 0.5700), VH)	((0.1190, 0.1688, 0.2356), A)	((0.1601, 0.2210, 0.3154), H)	((0.1432, 0.1886, 0.2483), VH)

To test the significance of the FIS, a two-sample t-test for the difference in means is conducted. The null hypothesis states that there is no significant difference between the FIS and the existing models (RSI, MACD, SO, OBV, and BH), with a significance level of 0.05. The alternative hypothesis suggests that the FIS outperforms the existing models. Using SPSS, the t-test is performed, and the results are presented in Table 6. From Table 6, the t-statistic value is −8.11606, and the p-value is $1.22 \times 10^{-8} (< .00001)$. The result is significant at $p \leq .05$, allowing us to reject the null hypothesis. Therefore, we accept the alternative hypothesis, indicating that the FIS performs better than RSI. Similarly, the results in Table 6 indicate that the FIS outperforms MACD, SO, and BH.

When comparing OBV and FI using a two-sample t-test, a t-statistic value of −0.6465 and a p-value of 0.2620 are obtained. The result is not significant, leading us to accept the null hypothesis. This implies that there is no significant difference between the returns of FI and OBV. Therefore, the findings of this study reject the weak form efficiency and provide evidence that profits can be achieved by utilizing past price and volume data.

8.2. Z-TOPSIS

To compare the results of the developed fuzzy indicator with the technical indicators, the Z number-based fuzzy TOPSIS method is employed. In order to construct the decision matrix, the expected return and risk values of each stock are calculated and presented in Table 4. Using these values, the average return is represented as a triangular fuzzy number. Additionally, the fuzzified risk-adjusted return (FSR, FSoR, FMR, and FAR) is computed for each alternative, as defined in Eqs. (9)–(12). These values are used to construct the decision matrix, which is presented in Table 7.

The weight of the criteria FSR calculated by fuzzy AHP is (0.3081, 0.4216, 0.5700). The reliability of the criteria given by the expert is very high (VH). So, the weight of the criteria FSR is denoted by the Z-number (0.3081, 0.4216, 0.5700, VH). Similarly, the remaining criteria weights and their reliabilities are given in Table 8.

Convert the reliability linguistic terms into their corresponding Z-number by using Table 2. Then, using step 5, the Z-numbers are transformed into a fuzzy number, which is given in Table 9.

Table 9
The fuzzy decision matrix with fuzzy weights.

	FSR	FSoR	FMR	FAR
RSI	(5.3183, 6.7547, 9.9043)	(5.7471, 9.1186, 16.5117)	(5.1969, 6.0115, 7.7978)	(5.0242, 5.1243, 5.3438)
MACD	(3.5131, 4.8734, 7.0292)	(0.4328, 4.6113, 11.2331)	(3.8893, 4.9229, 6.5607)	(4.8821, 4.9918, 5.1657)
SO	(3.1019, 4.9734, 6.9185)	(1.1837, 4.9465, 8.8573)	(4.1945, 5.0028, 5.8430)	(4.8866, 5.0004, 5.1187)
OBV	(7.7152, 10.0399, 11.4491)	(12.7114, 19.3137, 23.3159)	(5.8631, 6.5915, 7.0331)	(5.1427, 5.2631, 5.3361)
FI	(8.2598, 10.4344, 11.9257)	(8.5253, 10.8768, 12.4896)	(5.9049, 6.5009, 6.9096)	(5.1599, 5.2653, 5.3376)
weights	(0.2949,0.4036,0.5457)	(0.0841, 0.1193, 0.1665)	(0.1386, 0.1913, 0.2731)	(0.1370, 0.1805, 0.2377)

Table 10
The normalized fuzzy decision matrix.

	FSR	FSoR	FMR	FAR
RSI	(0.2687, 0.0344, 0.3649)	(0.2464, 0.0511, 0.3372)	(0.2979, 0.0267, 0.4258)	(0.0167, 0.0002, 0.0161)
MACD	(0.1775, 0.0326, 0.2527)	(0.0186, 0.0633, 0.1767)	(0.2229, 0.0339, 0.3424)	(0.0162, 0.0002 0.0156)
SO	(0.1567, 0.0449, 0.2402)	(0.0508, 0.0570, 0.1521)	(0.2404, 0.0265, 0.3289)	(0.0162 0.0002 0.0155)
OBV	(0.3898, 0.0557, 0.4594)	(0.5450, 0.1000, 0.5457)	(0.3361, 0.0239, 0.4226)	(0.0171 0.0002 0.0163)
FI	(0.4173, 0.0521, 0.4838)	(0.3655, 0.0356, 0.3183)	(0.3385, 0.0195, 0.4199)	(0.0171 0.0002 0.0163)
weights	(0.2949,0.4036,0.5457)	(0.0841, 0.1193, 0.1665)	(0.1386, 0.1913, 0.2731)	(0.1370, 0.1805, 0.2377)

Table 11
The weighted normalized fuzzy decision matrix.

	FSR	FSoR	FMR	FAR
RSI	(0.0792, 0.1047, 0.4534)	(0.0207, 0.0276, 0.1256)	(0.0413, 0.0556, 0.2567)	(0.0023, 0.0030, 0.0099)
MACD	(0.0523, 0.0681, 0.3093)	(0.0015, 0.0000, 0.0473)	(0.0309, 0.0409, 0.2019)	(0.0022, 0.0029, 0.0097)
SO	(0.0462, 0.0584, 0.2874)	(0.0043, 0.0040, 0.0466)	(0.0333, 0.0446, 0.2011)	(0.0022, 0.0029, 0.0097)
OBV	(0.1149, 0.1513, 0.5989)	(0.0458, 0.0615, 0.2275)	(0.0466, 0.0630, 0.2658)	(0.0023, 0.0031, 0.0102)
FI	(0.1231, 0.1629, 0.6345)	(0.0307, 0.0424, 0.1407)	(0.0469, 0.0637, 0.2654)	(0.0024, 0.0031, 0.0102)

Table 12
Fuzzy positive and negative ideal solution.

Alternative	RSI	MACD	SO	OBV	FI
s^+	0.1843	0.3532	0.3661	0.0225	0.0221
s^-	0.1847	0.0151	0.0053	0.3458	0.3645

Table 13
The preference order can be ranked indicator to the sort of p_i.

Alternative	RSI	MACD	SO	OBV	FI
R_i	0.5005	0.0411	0.0144	0.9387	0.9426
Rank	3	4	5	2	1

From this stage, we apply Jahanshahloo et al. [37] procedure. The normalized fuzzy decision matrix and the weighted normalized fuzzy decision matrix are calculated, which are shown in Table 10 and Table 11, respectively.

Next, the relative closeness of the alternative with respect to the positive ideal solution is calculated which is shown in Table 12.

Now, the set of alternatives is ranked through the descending order of the value of R_i, which is exhibited in Table 13. From Table 13, we observe that the proposed FI is getting the first rank. This shows that FI is better than the other technical indicators.

9. Limitations

Using a fuzzy inference system for stock price prediction comes with several limitations:

- The effectiveness of a fuzzy system heavily depends on the quality of the rule base. If the rules are not well-defined or are based on incorrect assumptions, the predictions can be unreliable.
- Fuzzy systems can generate complex rule sets that may be hard to interpret. Understanding and debugging the system can be challenging, especially for non-experts.
- Fuzzy systems may not handle outliers or extreme events well. Sudden, unexpected events (e.g., financial crises, geopolitical events) can significantly impact stock prices, and fuzzy systems might struggle to account for such occurrences.
- Fuzzy systems might not effectively account for market noise, rumors, or sentiment-driven fluctuations that can have a short-term impact on stock prices.
- Fuzzy systems might not adapt quickly to rapidly changing market conditions. They might lag behind in reacting to sudden shifts in trends.

It is important to note that no model, including fuzzy systems, can perfectly predict stock prices due to the inherent uncertainty and complexity of financial markets. Traders and investors should use such models as tools to inform their decisions, but always exercise caution and consider multiple sources of information.

10. Conclusion

The objective of this study was to evaluate the performance of technical indicators and fuzzy rule-based trading systems. Unlike previous studies, we ranked FI, MACD, RSI, SO, and OBV based on various evaluation metrics such as Sharpe ratio, Sortino ratio, Modigliani risk-adjusted performance (M^2), Value at Risk, standard deviation, and average return. Experimental data from the top 25 companies listed on the NASDAQ Exchange between 2015 and 2019 were collected for analysis.

The fuzzy inference system (FIS) was employed to generate trading signals (buy and sell) using the classical model output as input. The performance of the FIS approach was compared with the classical model output through statistical analysis and the Z-number-based fuzzy TOPSIS method. The results of both analyses demonstrated that the FIS approach outperformed the traditional technical indicators in trading simulations. This indicates that considering multiple technical trading outputs simultaneously can lead to improved prediction accuracy. Furthermore, the findings of this study highlight the profitability of the FIS approach as a trading strategy in the context of the NASDAQ stock exchange. It provides valuable insights for investors and researchers, enabling them to better understand market dynamics and the relationships between selected stocks. Additionally, the developed system can enhance the identification of price movement patterns and potential payouts. Its application by the USEC in scrutinizing investor portfolios can yield better results compared to the classical model.

However, it is important to note that the success of FIS-based stock trading depends on the quality of the fuzzy rules, the selection of appropriate input variables, and the effectiveness of the membership functions used. Proper validation and optimization of the FIS model are crucial to ensure its performance and reliability in real-world trading scenarios. The future research directions: Study the integration of FIS with other machine learning models (e.g., neural networks, reinforcement learning) to create hybrid systems that can capture complex relationships in the stock market. Extend the FIS to incorporate risk management strategies and portfolio optimization techniques, allowing for more sophisticated trading decisions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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